

Source Code

Turing Machine (TM) for $L = \{ a^n b^n c^n \mid n \geq 1 \}$ over $\{a, b, c\}$

```
def turing_machine(string: str) -> bool:
    tape = list(string)
    head = 0

    if not tape or any(ch not in "abc" for ch in tape):
        return False

    print(f"Input : {string}")
    print(f"Start : { ''.join(tape) }")
    pass_no = 0

    while True:
        head = 0
        while head < len(tape) and tape[head] == "X":
            head += 1

        if head == len(tape) or tape[head] != "a":
            break

        tape[head] = "X"

        head += 1
        while head < len(tape) and tape[head] in ("a", "X", "Y"):
            head += 1

        if head >= len(tape) or tape[head] != "b":
            return False
        tape[head] = "Y"

        head += 1
        while head < len(tape) and tape[head] in ("b", "Y", "Z"):
```

```
        head += 1
    if head >= len(tape) or tape[head] != "c":
        return False
    tape[head] = "Z"

    pass_no += 1
    print(f"Pass {pass_no}: {"".join(tape)}")

    if any(ch in "abc" for ch in tape):
        return False
    return True

string = input("Enter a string over {a, b, c}: ")

if turing_machine(string):
    print("\nAccepted")
else:
    print("\nRejected")
```

Output

```
Enter a string over {a, b, c}: aaabbbccc
Input : aaabbbccc
Start : aaabbbccc
Pass 1: XaaYbbZcc
Pass 2: XXaYYbZZc
Pass 3: XXXYYYZZZ

Accepted
(base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works>
```

```
Enter a string over {a, b, c}: aaabbbccca
Input : aaabbbccca
Start : aaabbbccca
Pass 1: XaaYbbZcca
Pass 2: XXaYYbZZca
Pass 3: XXXYYYZZZa

Rejected
(base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works>
```

Conclusion

The program successfully simulates a Turing Machine that verifies equal numbers of a, b, and c in the required order. It accepts strings belonging to $a^n b^n c^n$ and rejects strings with unequal counts or incorrect symbol ordering.